

T-S Fuzzy-Model-Based Robust H_∞ Design for Networked Control Systems With Uncertainties

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Abstract—This paper concerns a problem of robust H_∞ control for a class of uncertain nonlinear networked control systems (NCSs), which can be represented by a T-S fuzzy model with uncertainties. Both network-induced delay and packet dropout are addressed. The controller design method is presented based on a delay-dependent approach, and the robust H_∞ controller gain matrices are obtained by solving a set of linear matrix inequalities (LMIs). Moreover, a general Lyapunov–Krasovskii functional is used, and some slack matrices, which bring much flexibility in solving LMIs, are introduced during the proof. Simulation results show the validity of the proposed method.

Index Terms— H_∞ control, networked control system (NCS), norm bounded uncertainty, packet dropout, T-S model.

I. INTRODUCTION

NETWORKED control systems (NCSs) are systems where the feedback loop is closed via a communication network in which information, from various components such as sensors, controllers, and actuators, is exchanged with limited bandwidth. There are great benefits in NCSs such as reducing system wiring, ease of system diagnosis and maintenance, and increasing system agility, etc. However, despite the advantages and potentials, communication network in the control loops arise some new issues, especially the intermittent losses or delays of the communicated information due to use of a network, which imposes a challenge to system analysis and design. To address this challenge, many results have been developed in consideration of these two crucial issues, namely, network-induced delay and packet dropout [8], [12]–[16], [18], [19], [21]–[27], [29], [32], [33], [37]–[40], [42].

While some interesting techniques and results have been presented in the above-mentioned publications, the control of such systems still remains an open problem. For example, most of the control schemes mentioned above were developed

only focusing on linear NCSs, and nonlinear NCSs have received little attention, though some issues on nonlinear NCSs have been investigated such as the asymptotic behavior [33], input-to-state stability [23], input-to-output $\mathcal{L}_p$ stability with disturbances [24], and model-based method [21]. It should be noted that all these nonlinear results are for the stability analysis without addressing controller design. Furthermore, the precise knowledge of the NCSs are generally required except in [29] and [39]. However, the schemes in [29] and [39] suffer their own drawbacks. For example, the schemes developed in [29] and [39] are only for the linear NCSs.

It is well known that the T-S fuzzy model is qualified to represent a certain class of nonlinear dynamic systems, and many corresponding control techniques have been developed in the literature. The typical approach for controller design is carried out via the so-called parallel distributed compensation (PDC) method [34]. Moreover, many improved results have been proposed such as robust H_∞ control with norm-bounded uncertainties [31] or Frobenius norm-bounded uncertainties [20], and robust H_∞ static output feedback control [11]. Usually, the Lyapunov second method is used to study the stability of T-S fuzzy systems, where the majority of results are formulated as convex optimization techniques with linear matrix inequality (LMI) constraints. For a detailed review, please refer to [3], [11], [20], [30], [31], [34], [41], and references therein.

In this paper, we try to address the controller design problem for a class of nonlinear NCSs with parameter uncertainties and disturbances, where the nonlinear NCSs can be approximated by a T-S fuzzy model. In fact, fuzzy control technique has been applied to NCSs in [1], [14], and [45]. However, these results are very preliminary, and analytical studies are still missing, except [45], where a linear plant was modelled in discrete-time domain. To the best of the authors' knowledge, the controller design problem for nonlinear NCSs based on T-S fuzzy model has not been fully investigated, which motivates the present study.

In this research, the T-S fuzzy systems with norm-bounded uncertainties are utilized to characterize the nonlinear NCSs, and a robust H_∞ control scheme will be developed, where both network-induced delay and packet dropout are considered. There are several important features of the developed result that are worthy of mentioning. First, due to the sampling behavior, the control signal is not a continuous function as a result of the existence of the zero-order holder (ZOH) [6], but a piecewise constant function. Therefore, the nonlinear NCS is stabilized through piecewise signals with limited data instead of instantaneous signals. Second, in contrast to controller design methods in discrete-time domain, our method is formulated in the continuous-time domain, that is, the inter-sampling behavior is taken into account as well. Third, the controller design method is presented based on a delay-dependent approach and the robust H_∞ controller gain matrices are obtained by solving a set of LMIs, which is convenient for numerical solutions. Moreover, motivated by [9] and [17], a general Lyapunov–Krasovskii

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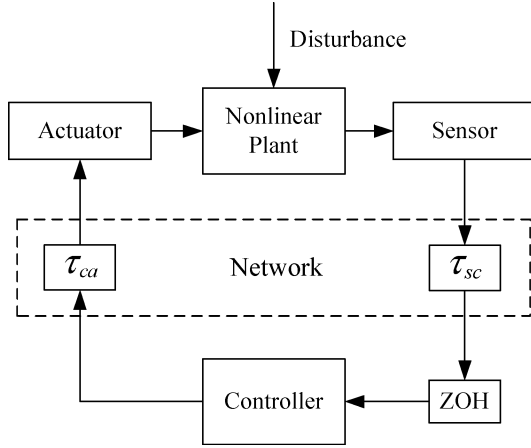


Fig. 1. Framework of networked control system.

functional is used and some slack matrices, which bring much flexibility in solving LMIs, are introduced during the proof of the main result. Finally, simulation examples are given to illustrate the effectiveness of the proposed method.

Before presenting the results, some notations are required. Throughout this paper, the superscript T stands for matrix transposition, $\mathbb{R}^n$ denotes the n -dimensional Euclidean space, $\mathbb{R}^{n \times m}$ is the set of all $n \times m$ real matrices, and the notation $X > 0$ (respectively, $X \geq 0$), for $X \in \mathbb{R}^{n \times n}$, means that X is symmetric and positive definite (respectively, positive semidefinite). Identity and zero matrices, of appropriate dimensions, will be denoted by I and 0 , respectively. The notation $*$ always denotes the symmetric block in one symmetric matrix. $\mathcal{L}_2$ is the space of square integrable functions over $[0, \infty)$, and $\|\cdot\|_2$ denotes the $\mathcal{L}_2$ -norm.

II. SYSTEM DESCRIPTION

A typical NCS model with network-induced delays is shown in Fig. 1, where τ_{sc} is sensor-to-controller delay and τ_{ca} is the controller-to-actuator delay. It is assumed that the controller computational delay can be absorbed into either τ_{sc} or τ_{ca} .

The plant in Fig. 1 is a nonlinear plant, which can be described by the following T-S fuzzy model:

Plant rule i :

IF $\theta_1(t)$ is F_{i1} , $\dots$, and $\theta_s(t)$ is F_{is}

THEN

$$\dot{x}(t) = (A_i + \Delta A_i)x(t) + (B_i + \Delta B_i)u(t) + B_{1i}w(t)$$

$$z(t) = C_i x(t) + D_i u(t) \quad \text{for } i = 1, 2, \dots, r \quad (1)$$

where $x(t) \in \mathbb{R}^n$ is the state vector; $u(t) \in \mathbb{R}^m$ is the control input vector; $w(t) \in \mathcal{L}_2$ is the external disturbance and/or modeling error signal; $z(t) \in \mathbb{R}^{n_z}$ is the controlled output; r and n are the numbers of rules and state variables, respectively; A_i , B_i , B_{1i} , C_i , and D_i are constant matrices with appropriate dimensions; F_{ig} is the fuzzy set ($g = 1, 2, \dots, s$); and $\theta_1(t), \theta_2(t), \dots, \theta_s(t)$ are the premise variables. The matrices ΔA_i and ΔB_i denote the norm-bounded uncertainties in system, and they are of the form

$$[\Delta A_i, \Delta B_i] = GF(t)[E_{ai}, E_{bi}] \quad (2)$$

where G , E_{ai} , and E_{bi} are known constant matrices of appropriate dimensions, and $F(t)$ is a time-varying matrix with Lebesgue-measurable elements satisfying $F^T(t)F(t) \leq I$.

By using product inference, singleton fuzzifier, and center-average defuzzifier, the inferred system can be expressed by [31]

$$\begin{aligned} (\Sigma_o) : \dot{x}(t) &= \sum_{i=1}^r h_i(\theta(t)) [(A_i + \Delta A_i)x(t) \\ &\quad + (B_i + \Delta B_i)u(t) + B_{1i}w(t)] \\ z(t) &= \sum_{i=1}^r h_i(\theta(t)) [C_i x(t) + D_i u(t)] \end{aligned} \quad (3)$$

where

$$h_i(\theta(t)) = \frac{\mu_i(\theta(t))}{\sum_{i=1}^r \mu_i(\theta(t))}, \quad \mu_i(\theta(t)) = \prod_{g=1}^s F_{ig}(\theta_g(t))$$

and $F_{ig}(\theta_g(t))$ is the grade of membership function F_{ig} .

Usually, it is assumed that $1 \geq \mu_i(\theta(t)) \geq 0$, and $\sum_{i=1}^r \mu_i(\theta(t)) > 0$, for all t . Then, we can see that $h_i(\theta(t)) \geq 0$, for $i = 1, 2, \dots, r$, and $\sum_{i=1}^r h_i(\theta(t)) = 1$.

Due to the sampling behavior of an NCS, the control signal is not a continuous function as a result of the existence of a ZOH, but a piecewise constant function. So the nonlinear NCS can be stabilized by piecewise signal with limited data instead of instantaneous signal. As seen from Fig. 1, the network-induced delay consists of sensor-to-controller delay τ_{sc} and the controller-to-actuator delay τ_{ca} . Let $t_k, t_{k+1}, \dots, t_{k+q}$ ($k = 0, 1, 2, 3, \dots$) be the sampling instants, $h = t_{k+1} - t_k$ be the sampling period, and $\tau_k, \tau_{k+1}, \dots, \tau_{k+q}$ be corresponding network-induced delays, respectively, where q is a positive integer. Note that τ_k is a combination of τ_{sc} and τ_{ca} , and the controller computational delay are absorbed into either τ_{sc} or τ_{ca} , and $\tau_0 = 0$. In addition, τ_k needs not to be restricted within one sampling period. Thus, the control signals may be transmitted to the plant at the instant $t_k + \tau_k, t_{k+1} + \tau_{k+1}, \dots, t_{k+q} + \tau_{k+q}$.

Two definitions are often used when developing the controller for an NCS. One is maximum allowable delay bound η , which is defined as the maximum allowable interval from the instant when sensor nodes sample sensor data from a plant, to the instant when actuators output the transferred data to the plant [13]. The other is maximum allowable transfer interval ζ , which is defined as a deadline if a transmission of control data takes place at time $t_k + \tau_k$, then another one must occur within the time interval $[t_k + \tau_k, t_k + \tau_k + \zeta)$ [33]. In this study, maximum allowable control interval is defined as $\delta = \eta + \zeta$ by combining the above two definitions, which is used to analyze the network-induced delay and packet dropout problems (see Fig. 2).

In feedback control systems, it is important that sampled data should be transmitted within a sampling period and that stability of control systems should be guaranteed. While a shorter sampling period is preferable in most control systems, for some purposes, it can be lengthened up to a certain bound within which stability of the system is guaranteed in spite of the performance degradation. Consider the situation that two sampling data packets arrive at the controller at instant $t_k + \tau_k$ and $t_{k+q} + \tau_{k+q}$ in sequence.

If $q = 1$, no packet dropout occurs during the time interval $[t_k, t_{k+1} + \tau_{k+1})$. If $q > 1$, $q - 1$ packets lost during the time interval $[t_k, t_{k+q} + \tau_{k+q})$.

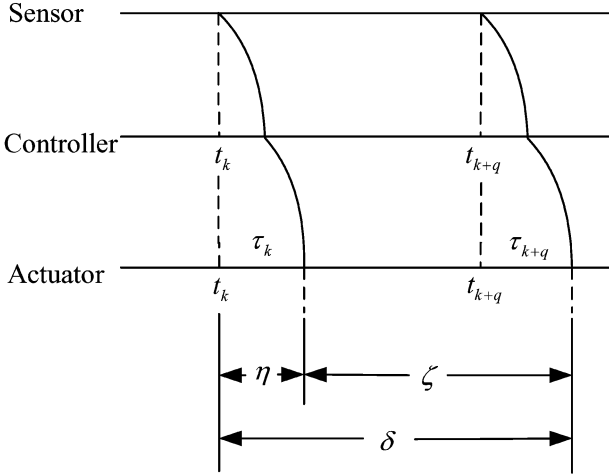


Fig. 2. Time-sequence diagram of two sequential control signals in the NCS.

For given δ , the following inequality should hold so that the stability of closed-loop system may be guaranteed:

$$t_{k+q} + \tau_{k+q} - t_k < \delta. \quad (4)$$

Before further discussions, the following assumptions are required.

Assumption 1: The sensor is time-driven. The controller and the actuator are event-driven. The clocks among them are synchronized. The packets reach the controller and actuator by their original transmitting sequence if they are not lost.

Many protocols satisfy the above assumption, such as TCP/IP at the application level and IEEE 802.5 for token ring [45].

Assumption 2: There exists a maximum bound of network-induced delay in the NCS, that is, $\tau_{max} \geq \tau_k$, ($k = 0, 1, 2, \dots$).

The maximal value of q can be derived as follows:

$$q_{max} = \text{nint} \left[\frac{\delta - \tau_{max}}{h} \right] \quad (5)$$

where $\text{nint}[\cdot]$ denotes the nearest integer part of $[\cdot]$.

Further, the maximal allowable packet dropout rate r_{max} is defined as follows:

$$r_{max} = \frac{q_{max} - 1}{q_{max}}. \quad (6)$$

Remark 1: It is explicit that $r_{max} = 0$ means any packet dropout will make the system unstable. In fact, the motive of (5) and (6) is used to choose proper sampling period h for a given δ based on the real network condition. For example, if the actual network packet dropout rate is r that can be measured by experiments, we should choose r_{max} satisfying $r_{max} > r$ by (6). Then, using this q_{max} , the sampling period h can thus be obtained from (5), which in turn implies the condition (4).

For given δ , the smaller h will lead to the larger q_{max} , the allowable packet dropout rate may be higher, but the amount of communication will be required much. However, the larger h will lead to the lower allowable packet dropout rate, which may degrade the performance of the system. The relationship between δ and the performance of NCSs will be analyzed in next section in detail.

Now we design the controller as the following form in order to stabilize the closed-loop system:

Control rule i :

IF $\theta_1(t_k)$ is F_{i1} , $\dots$, and $\theta_s(t_k)$ is F_{is}

THEN $u(t) = K_i x(t_k)$, for $i = 1, 2, \dots, r$. (7)

Hence, the inferred fuzzy controller is given by

$$u(t) = \sum_{i=1}^r h_i(\theta(t_k)) K_i x(t_k), \quad t \in [t_k + \tau_k, t_{k+q} + \tau_{k+q}] \quad (8)$$

where K_i ($i = 1, 2, \dots, r$) is the fuzzy control gain matrix, and $x(t_k)$ is the state vector of plant at the instant t_k ($k = 0, 1, 2, 3, \dots$). In this study, we assume that $u(t) = 0$ before the first control signal reaches the plant.

In the presence of network, network-induced delay and packet dropout occur inevitably, which may degrade the system dynamic performance, and even may cause instability.

In this study, our aim is to develop a method for designing state feedback fuzzy controllers (8) for the system Σ_o in network-based environment such that the following requirements are satisfied.

- 1) The closed-loop system is asymptotically stable for all admissible uncertainties.
- 2) Under zero initial condition, the closed-loop system satisfies H_∞ performance, i.e., $\|z(t)\|_2 < \gamma \|w(t)\|_2$, for any nonzero $w(t) \in \mathcal{L}_2$, where $\gamma > 0$ denotes prescribed attenuation level.

For simplicity, the following notations are used: $x(t) = x$, $x(t_k) = x_k$, $w(t) = w$, $z(t) = z$, $h_i(\theta(t)) = h_i$, and $h_i(\theta(t_k)) = h_i^k$. Then the closed-loop system can be expressed as

$$\begin{aligned} (\Sigma_c) : \dot{x} &= \sum_{i=1}^r \sum_{j=1}^r h_i h_j^k [(A_i + \Delta A_i)x \\ &\quad + (B_i + \Delta B_i)K_j x_k + B_{1i}w] \\ z &= \sum_{i=1}^r \sum_{j=1}^r h_i h_j^k (C_i x + D_i K_j x_k) \\ t &\in [t_k + \tau_k, t_{k+q} + \tau_{k+q}]. \end{aligned} \quad (9)$$

III. MAIN RESULTS

To obtain our results, we need the following fundamental lemmas.

Lemma 1 ([7]): For any real matrices X_i, Y_i , for $1 \leq i \leq r$, and $S > 0$ with appropriate dimensions, we have

$$\begin{aligned} &2 \sum_{i=1}^r \sum_{j=1}^r \sum_{p=1}^r \sum_{l=1}^r h_i h_j h_p h_l X_{ij}^T S Y_{pl} \\ &\leq \sum_{i=1}^r \sum_{j=1}^r h_i h_j (X_{ij}^T S X_{ij} + Y_{ij}^T S Y_{ij}). \end{aligned} \quad (10)$$

Lemma 2: For any real matrix $Y > 0$ with appropriate dimensions, the following inequalities hold for $t \in [t_k + \tau_k, t_{k+q} + \tau_{k+q}]$:

$$\begin{aligned} &\int_{t_k}^t x^T(s) Y x(s) ds < \int_{t-\delta}^t x^T(s) Y x(s) ds \\ &\int_{t_k}^t \dot{x}^T(s) Y \dot{x}(s) ds < \int_{t-\delta}^t \dot{x}^T(s) Y \dot{x}(s) ds. \end{aligned} \quad (11)$$

Proof: For $t \in [t_k + \tau_k, t_{k+q} + \tau_{k+q})$ and the condition (4), we have $\tau_k \leq t - t_k < \delta$. Since the integrand is positive definite, (11) holds obviously. ■

Lemma 3 ([35]): Given matrices $X = X^T, G, E$, and $Y > 0$ with appropriate dimensions

$$X + GF(t)E + E^T F^T(t)G^T < 0 \quad (12)$$

for all $F(t)$ satisfying $F^T(t)F(t) \leq Y$, if and only if there exists a scalar $\varepsilon > 0$ such that

$$X + \varepsilon^{-1}GG^T + \varepsilon E^T Y E < 0. \quad (13)$$

The design objective in this study is to specify the controller in (8) for the system Σ_o in network-based environment such that the requirements 1) and 2) are satisfied. Then we propose the following results.

Theorem 1: Given scalars $\delta > 0, \gamma > 0$, and ϵ_l ($l = 1, \dots, 4$), if there exist matrices $P > 0, Q_1 > 0, Q_3 > 0, Q_2, N_g$ ($g = 1, \dots, 3$), X_j ($j = 1, \dots, r$), and Y with appropriate dimensions, as well as the scalars $\varepsilon_{ij} > 0$, such that the LMIs hold in (14) at the bottom of the page, for any $1 \leq i \leq j \leq r$, where

$$\begin{aligned} \Xi_{11} &= \delta Q_1 + N_1 + N_1^T - \epsilon_1 A_i Y - \epsilon_1 Y^T A_i^T \\ \Xi_{12} &= -N_1 + N_2^T - \epsilon_1 B_i X_j - \epsilon_2 Y^T A_i^T \\ \Xi_{13} &= P + \delta Q_2 + N_3^T + \epsilon_1 Y - \epsilon_3 Y^T A_i^T \\ \Xi_{14} &= -\epsilon_1 B_{1i} \\ \Xi_{22} &= -N_2 - N_2^T - \epsilon_2 B_i X_j - \epsilon_2 X_j^T B_i^T \\ \Xi_{23} &= -N_3^T + \epsilon_2 Y - \epsilon_3 X_j^T B_i^T \\ \Xi_{24} &= -\epsilon_2 B_{1i} \\ \Xi_{33} &= \delta Q_3 + \epsilon_3 Y + \epsilon_3 Y^T \\ \Xi_{34} &= -\epsilon_3 B_{1i} \\ \Xi_{44} &= -\gamma^2 I \end{aligned}$$

then the system Σ_c is asymptotically stable with an H_∞ performance for the prescribed γ . Moreover, the controller gain matrices can be obtained as $K_j = X_j Y^{-1}$.

Proof: Since (14) implies that $\begin{bmatrix} Q_1 & Q_2 \\ Q_2^T & Q_3 \end{bmatrix} > 0$, we can choose the Lyapunov–Krasovskii functional as

$$V(t) = x^T P x + \int_{-\delta}^0 \int_{t+v}^t \begin{bmatrix} x(s) \\ \dot{x}(s) \end{bmatrix}^T \begin{bmatrix} Q_1 & Q_2 \\ Q_2^T & Q_3 \end{bmatrix} \begin{bmatrix} x(s) \\ \dot{x}(s) \end{bmatrix} ds dv. \quad (15)$$

For any matrices N_g ($g = 1, \dots, 3$) of appropriate dimensions, it can be seen that, from the Newton–Leibniz formula

$$2 \left[x^T N_1 + x_k^T N_2 + \dot{x}^T N_3 \right] \times \left[x - x_k - \int_{t_k}^t \dot{x}(s) ds \right] = 0. \quad (16)$$

Moreover, from the system equation in (9), we have

$$\begin{aligned} & 2 \left[x^T M_1 + x_k^T M_2 + \dot{x}^T M_3 + \int_{t_k}^t x(s) ds M_4 \right] \\ & \times \left[\dot{x} + \sum_{i=1}^r \sum_{j=1}^r h_i h_j^k (-A_i + \Delta A_i) x \right. \\ & \left. - (B_i + \Delta B_i) K_j x_k - B_{1i} w \right] = 0 \end{aligned} \quad (17)$$

for any matrices M_l ($l = 1, \dots, 4$) of appropriate dimensions.

It also follows that from Lemma 1

$$\begin{aligned} z^T z &= \sum_{i=1}^r \sum_{j=1}^r \sum_{p=1}^r \sum_{l=1}^r h_i h_j^k h_p h_l^k (C_i x + D_i K_j x_k)^T \\ & \times (C_p x + D_p K_l x_k) \\ & \leq \sum_{i=1}^r \sum_{j=1}^r h_i h_j^k (C_i x + D_i K_j x_k)^T \\ & \times (C_i x + D_i K_j x_k). \end{aligned} \quad (18)$$

Considering (16)–(18) and Lemma 2, along the trajectories of system (9), the corresponding time derivative of $V(t)$, for $t \in [t_k + \tau_k, t_{k+q} + \tau_{k+q})$, is given by

$$\begin{aligned} \dot{V}(t) &= 2x^T P \dot{x} + \delta \begin{bmatrix} x \\ \dot{x} \end{bmatrix}^T \begin{bmatrix} Q_1 & Q_2 \\ Q_2^T & Q_3 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \end{bmatrix} \\ & - \int_{t-\delta}^t \begin{bmatrix} x(s) \\ \dot{x}(s) \end{bmatrix}^T \begin{bmatrix} Q_1 & Q_2 \\ Q_2^T & Q_3 \end{bmatrix} \begin{bmatrix} x(s) \\ \dot{x}(s) \end{bmatrix} ds \\ & + 2 \left[x^T N_1 + x_k^T N_2 + \dot{x}^T N_3 \right] \\ & \times \left[x - x_k - \int_{t_k}^t \dot{x}(s) ds \right] \\ & + 2 \left[x^T M_1 + x_k^T M_2 + \dot{x}^T M_3 + \int_{t_k}^t x(s) ds M_4 \right] \end{aligned}$$

$$\Xi^{(ij)} := \begin{bmatrix} \Xi_{11} & \Xi_{12} & \Xi_{13} & \Xi_{14} & \epsilon_4 \delta Y^T A_i^T & \delta N_1 & Y^T C_i^T & -\epsilon_1 G & \varepsilon_{ij} Y^T E_{a_i}^T \\ * & \Xi_{22} & \Xi_{23} & \Xi_{24} & \epsilon_4 \delta X_j^T B_i^T & \delta N_2 & X_j^T D_i^T & -\epsilon_2 G & \varepsilon_{ij} X_j^T E_{b_i}^T \\ * & * & \Xi_{33} & \Xi_{34} & -\epsilon_4 \delta Y^T & \delta N_3 & 0 & -\epsilon_3 G & 0 \\ * & * & * & \Xi_{44} & \epsilon_4 \delta B_{1i}^T & 0 & 0 & 0 & 0 \\ * & * & * & * & -\delta Q_1 & -\delta Q_2 & 0 & \epsilon_4 \delta G & 0 \\ * & * & * & * & * & -\delta Q_3 & 0 & 0 & 0 \\ * & * & * & * & * & * & -I & 0 & 0 \\ * & * & * & * & * & * & * & -\varepsilon_{ij} I & 0 \\ * & * & * & * & * & * & * & * & -\varepsilon_{ij} I \end{bmatrix} < 0 \quad (14)$$

$$\begin{aligned}
& \times \left[\dot{x} + \sum_{i=1}^r \sum_{j=1}^r h_i h_j^k (-(A_i + \Delta A_i)x \right. \\
& \quad \left. - (B_i + \Delta B_i)K_j x_k - B_{1i}w) \right] \\
& + \delta \alpha^T(t) \tilde{N} \tilde{Q}^{-1} \tilde{N}^T \alpha(t) \\
& - \int_{t-\delta}^t \alpha^T(t) \tilde{N} \tilde{Q}^{-1} \tilde{N}^T \alpha(t) ds \\
& + z^T z - \gamma^2 w^T w - z^T z + \gamma^2 w^T w \\
& < \sum_{i=1}^r \sum_{j=1}^r h_i h_j^k \left[\alpha^T(t) (\tilde{\Phi} + \delta \tilde{N} \tilde{Q}^{-1} \tilde{N}^T) \alpha(t) \right. \\
& \quad \left. - \int_{t_k}^t [\alpha^T(t) \tilde{N} + \beta^T(s) \tilde{Q}] \tilde{Q}^{-1} \right. \\
& \quad \left. \times [\alpha^T(t) \tilde{N} + \beta^T(s) \tilde{Q}]^T ds \right] \\
& - z^T z + \gamma^2 w^T w
\end{aligned} \tag{19}$$

where

$$\begin{aligned}
\alpha^T(t) &= [x^T, x_k^T, \dot{x}^T, w^T], \quad \beta^T(s) = [x^T(s), \dot{x}^T(s)] \\
\tilde{\Phi} &= \begin{bmatrix} \Phi_{11} + C_i^T C_i & \Phi_{12} + C_i^T D_i K_j & \Phi_{13} & \Phi_{14} \\ * & \Phi_{22} + (D_i K_j)^T (D_i K_j) & \Phi_{23} & \Phi_{24} \\ * & * & \Phi_{33} & \Phi_{34} \\ * & * & * & \Phi_{44} \end{bmatrix} \\
\tilde{N} &= \begin{bmatrix} (A_i + \Delta A_i)^T M_4^T & N_1 \\ K_j^T (B_i + \Delta B_i)^T M_4^T & N_2 \\ -M_4^T & N_3 \\ B_{1i}^T M_4^T & 0 \end{bmatrix}, \quad \tilde{Q} = \begin{bmatrix} Q_1 & Q_2 \\ Q_2^T & Q_3 \end{bmatrix} \\
\Phi_{11} &= \delta Q_1 + N_1 + N_1^T - M_1(A_i + \Delta A_i) \\
&\quad - (A_i + \Delta A_i)^T M_1^T \\
\Phi_{12} &= -N_1 + N_2^T - M_1(B_i + \Delta B_i)K_j \\
&\quad - (A_i + \Delta A_i)^T M_2^T \\
\Phi_{13} &= P + \delta Q_2 + N_3^T + M_1 - (A_i + \Delta A_i)^T M_3^T \\
\Phi_{14} &= -M_1 B_{1i} \\
\Phi_{22} &= -N_2 - N_2^T - M_2(B_i + \Delta B_i)K_j \\
&\quad - K_j^T (B_i + \Delta B_i)^T M_2^T \\
\Phi_{23} &= -N_3^T + M_2 - K_j^T (B_i + \Delta B_i)^T M_3^T \\
\Phi_{24} &= -M_2 B_{1i} \\
\Phi_{33} &= \delta Q_3 + M_3 + M_3^T \\
\Phi_{34} &= -M_3 B_{1i} \\
\Phi_{44} &= -\gamma^2 I.
\end{aligned}$$

Using the Schur complement [2], we know $\tilde{\Phi} + \delta \tilde{N} \tilde{Q}^{-1} \tilde{N}^T < 0$ is equivalent to (20) at the bottom of the page, for any $1 \leq i \leq j \leq r$.

For convenience, we use $\Phi_o^{(ij)}$ to denote $\Phi^{(ij)}$ without uncertainties, i.e., $\Delta A_i = 0, \Delta B_i = 0$ in (20). Then, we have $\Phi^{(ij)} = \Phi_o^{(ij)} + \Delta \Phi^{(ij)}$, where

$$\begin{aligned}
\Delta \Phi^{(ij)} &= \tilde{M}^T [\Delta A_i \quad \Delta B_i K_j \quad 0 \quad 0 \quad 0 \quad 0] \\
&\quad + [\Delta A_i \quad \Delta B_i K_j \quad 0 \quad 0 \quad 0 \quad 0]^T \tilde{M}, \\
\tilde{M} &= [-M_1^T \quad -M_2^T \quad -M_3^T \quad 0 \quad \delta M_4^T \quad 0 \quad 0].
\end{aligned} \tag{21}$$

Considering the parameter uncertainties are satisfied (2), we can obtain

$$\Phi_o^{(ij)} + \mathcal{G}^T F(t) \mathcal{E} + \mathcal{E}^T F^T(t) \mathcal{G} < 0 \tag{22}$$

where

$$\begin{aligned}
\mathcal{G} &= [-G^T M_1^T \quad -G^T M_2^T \quad -G^T M_3^T \quad 0 \quad \delta G^T M_4^T \quad 0 \quad 0] \\
\mathcal{E} &= [E_{ai} \quad E_{bi} K_j \quad 0 \quad 0 \quad 0 \quad 0].
\end{aligned}$$

Utilizing Lemma 3, (22) implies that there exists scalars $\varepsilon_{ij} > 0$ such that

$$\Phi_o^{(ij)} + \varepsilon_{ij}^{-1} \mathcal{G}^T \mathcal{G} + \varepsilon_{ij} \mathcal{E}^T \mathcal{E} < 0. \tag{23}$$

Applying the Schur complement again, it can be concluded that (23) is equivalent to

$$\Psi^{(ij)} := \begin{bmatrix} \Phi_o^{(ij)} & \mathcal{G}^T & \varepsilon_{ij} \mathcal{E}^T \\ * & -\varepsilon_{ij} I & 0 \\ * & * & -\varepsilon_{ij} I \end{bmatrix} < 0 \tag{24}$$

for any $1 \leq i \leq j \leq r$.

Set $M_1 = \epsilon_1 M$, $M_2 = \epsilon_2 M$, $M_3 = \epsilon_3 M$, and $M_4 = \epsilon_4 M$. Thus, $\Xi^{(ij)} < 0$ implies that M is nonsingular since Ξ_{33} in (14) must be negative definite. Then, pre, post-multiplying both sides of $\Psi^{(ij)}$ with $\text{diag}(M^{-1}, M^{-1}, M^{-1}, I, M^{-1}, M^{-1}, I, I, I)$ and its transpose, respectively, setting $Y = M^{-T}$, $X_j = K_j Y$, and replacing $M^{-1} P M^{-T}$, $M^{-1} Q_1 M^{-T}$, $M^{-1} Q_2 M^{-T}$, $M^{-1} Q_3 M^{-T}$, and $M^{-1} N_g M^{-T}$ with P , Q_1 , Q_2 , Q_3 , and N_g ($g = 1, \dots, 3$), respectively, we can obtain (14) directly, which implies

$$\dot{V}(t) < -z^T z + \gamma^2 w^T w \tag{25}$$

for $t \in [t_k + \tau_k, t_{k+q} + \tau_{k+q})$.

Integrating both sides of (25) from $t_k + \tau_k$ to $t \in [t_k + \tau_k, t_{k+q} + \tau_{k+q})$, we get

$$\begin{aligned}
V(t) - V(t_k + \tau_k) &< - \int_{t_k + \tau_k}^t z^T(t) z(t) dt \\
&\quad + \int_{t_k + \tau_k}^t \gamma^2 w^T(t) w(t) dt.
\end{aligned} \tag{26}$$

$$\Phi^{(ij)} := \begin{bmatrix} \Phi_{11} & \Phi_{12} & \Phi_{13} & \Phi_{14} & \delta(A_i + \Delta A_i)^T M_4^T & \delta N_1 & C_i^T \\ * & \Phi_{22} & \Phi_{23} & \Phi_{24} & \delta K_j^T (B_i + \Delta B_i)^T M_4^T & \delta N_2 & K_j^T D_i^T \\ * & * & \Phi_{33} & \Phi_{34} & -\delta M_4^T & \delta N_3 & 0 \\ * & * & * & \Phi_{44} & \delta B_{1i}^T M_4^T & 0 & 0 \\ * & * & * & * & -\delta Q_1 & -\delta Q_2 & 0 \\ * & * & * & * & * & -\delta Q_3 & 0 \\ * & * & * & * & * & * & -I \end{bmatrix} < 0 \tag{20}$$

Since $V(t)$ is continuous in $t \in [t_0, \infty)$, it can be seen explicitly that

$$V(t) - V(t_0) < - \int_{t_0}^t z^T(t)z(t)dt + \int_{t_0}^t \gamma^2 w^T(t)w(t)dt. \quad (27)$$

Letting $t \rightarrow \infty$ and under zero initial condition, the H_∞ performance can be satisfied, i.e., $\|z\|_2 < \gamma\|w\|_2$, for any nonzero $w(t) \in \mathcal{L}_2$. ■

Remark 2: Theorem 1 provides a delay-dependent criterion on the robust H_∞ control for system Σ_o in terms of the solvability of a set of LMIs. It is worth pointing out for a given scalar $\gamma > 0$, the minimum $1/\delta$, which satisfies the LMIs in (14), can be computed by solving a generalized eigenvalue minimization problem, i.e.,

$$\begin{aligned} & \text{minimize } \frac{1}{\delta} \\ & \text{subject to } P > 0, Q_1 > 0, Q_3 > 0, (14), \text{ for a given } \gamma. \end{aligned} \quad (28)$$

In the same manner, for a given $\delta > 0$, the minimum γ , which satisfies the LMIs in (14), can also be computed by solving a linear objective minimization problem, i.e.,

$$\begin{aligned} & \text{minimize } \gamma \\ & \text{subject to } P > 0, Q_1 > 0, Q_3 > 0, (14), \text{ for a given } \delta. \end{aligned} \quad (29)$$

To this end, there exists a tradeoff between δ and γ , which are related to the maximum bound of network-induced delay and the maximal allowable packet dropout rate, and the performance of the NCS, respectively.

Remark 3: The optimal values of the tuning parameters ϵ_l ($l = 1, \dots, 4$) that were introduced in Theorem 1 can be found by the method stated in [4] and [43]. We choose the cost function t_{min} , which is obtained by solving the feasibility problem using Matlab's LMI Toolbox (The MathWorks, Version 1.0.8, 1995). There exists no feasible solution to the set of LMIs under consideration, when t_{min} is positive. Applying a numerical optimization algorithm, such as **fminunc** in the Optimization Toolbox (The MathWorks, Version 3, 2004), to the above cost function, a locally convergent solution to the problem is obtained. If the resulting minimum value of the cost function is negative, the tuning parameters that solve the problem can be obtained.

When there is no parameter uncertainties in the system Σ_c , Theorem 1 is specialized as follows.

Corollary 1: Given scalars $\delta > 0$, $\gamma > 0$, and ϵ_l ($l = 1, \dots, 4$), if there exist matrices $P > 0$, $Q_1 > 0$, $Q_3 > 0$, Q_2 , N_g ($g = 1, \dots, 3$), X_j ($j = 1, \dots, r$), and Y with appropriate dimensions, as well as the scalars $\epsilon_{ij} > 0$, such that the following LMIs hold:

$$\begin{bmatrix} \Xi_{11} & \Xi_{12} & \Xi_{13} & \Xi_{14} & \epsilon_4 \delta Y^T A_i^T & \delta N_1 & Y^T C_i^T \\ * & \Xi_{22} & \Xi_{23} & \Xi_{24} & \epsilon_4 \delta X_j^T B_i^T & \delta N_2 & X_j^T D_i^T \\ * & * & \Xi_{33} & \Xi_{34} & -\epsilon_4 \delta Y^T & \delta N_3 & 0 \\ * & * & * & \Xi_{44} & \epsilon_4 \delta B_{1i}^T & 0 & 0 \\ * & * & * & * & -\delta Q_1 & -\delta Q_2 & 0 \\ * & * & * & * & * & -\delta Q_3 & 0 \\ * & * & * & * & * & * & -I \end{bmatrix} < 0 \quad (30)$$

for any $1 \leq i \leq j \leq r$, where $\Xi_{i'j'}$ ($1 \leq i' \leq j' \leq 4$) are defined in (14), the system Σ_c with $\Delta A_i = 0$, $\Delta B_i = 0$ is asymptotically stable with an H_∞ performance for the prescribed γ . Moreover, the controller gain matrices are $K_j = X_j Y^{-1}$.

The result of Theorem 1 can be further improved by using a relaxed LMI method as follows.

Theorem 2: Given scalars $\delta > 0$, $\gamma > 0$, and ϵ_l ($l = 1, \dots, 4$), if there exist matrices $P > 0$, $Q_1 > 0$, $Q_3 > 0$, Q_2 , N_g ($g = 1, \dots, 3$), Y , X_j , Z_{ij} , $Z_{ii} = Z_{ii}^T$, ($i = 1, \dots, r$, $i < j \leq r$) with appropriate dimensions, as well as the scalars $\epsilon_{ij} > 0$, such that the following LMIs hold:

$$\Xi^{(ii)} < Z_{ii}, \quad i = 1, \dots, r \quad (31)$$

$$\Xi^{(ij)} + \Xi^{(ji)} \leq Z_{ij} + Z_{ij}^T, \quad i = 1, \dots, r, i < j \leq r \quad (32)$$

$$\begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1r} \\ * & Z_{22} & \cdots & Z_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & Z_{rr} \end{bmatrix} \leq 0 \quad (33)$$

where $\Xi^{(ij)}$ are defined in (14), the system Σ_c is asymptotically stable with an H_∞ performance for the prescribed γ . Moreover, the controller gain matrices are $K_j = X_j Y^{-1}$.

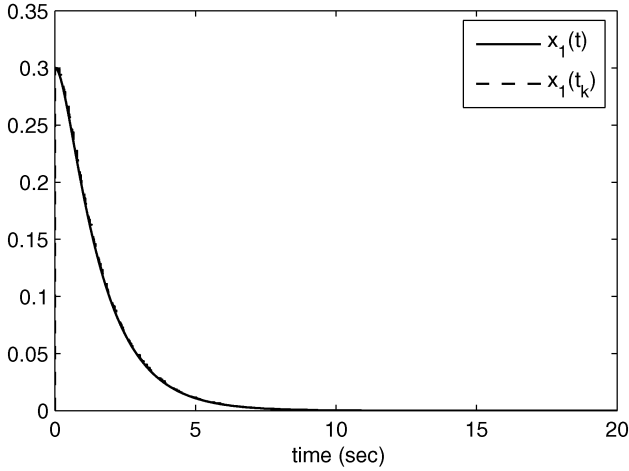
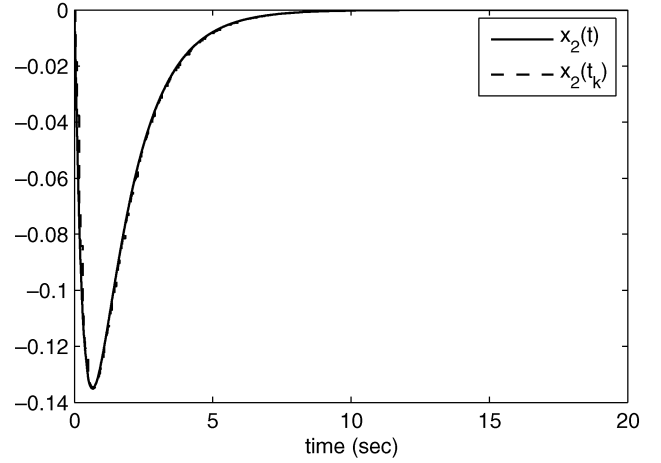
Proof: Using (31)–(33), we have

$$\begin{aligned} \sum_{i=1}^r \sum_{j=1}^r h_i h_j^k \Xi^{(ij)} &= \sum_{i=1}^r h_i h_i^k \Xi^{(ii)} \\ &+ \sum_{i=1}^{r-1} \sum_{j=i+1}^r h_i h_j^k (\Xi^{(ij)} + \Xi^{(ji)}) \\ &< \sum_{i=1}^r h_i h_i^k Z_{ii} \\ &+ \sum_{i=1}^{r-1} \sum_{j=i+1}^r h_i h_j^k (Z_{ij} + Z_{ji}^T) \\ &= \begin{bmatrix} h_1 I \\ h_2 I \\ \vdots \\ h_r I \end{bmatrix}^T \begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1r} \\ * & Z_{22} & \cdots & Z_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ * & * & \cdots & Z_{rr} \end{bmatrix} \\ &\times \begin{bmatrix} h_1^k I \\ h_2^k I \\ \vdots \\ h_r^k I \end{bmatrix} \\ &\leq 0. \end{aligned} \quad (34)$$

From inequality (34), we know $\sum_{i=1}^r \sum_{j=1}^r h_i h_j^k \Xi^{(ij)} < 0$ if (31)–(33) are satisfied, which implies that (14) holds. ■

IV. SIMULATION

To illustrate the developed H_∞ controller design approach for the NCSs, we consider three examples. The first is a mass-spring system that can be expressed precisely by a T-S fuzzy system if not considering external disturbance. The second is a single-link rigid robot system. The third is a continuous stirred tank reactor (CSTR) that is a classical industrial plant.


 Fig. 3. Trajectories of $x_1(t)$ and $x_1(t_k)$ with $h = 0.02s$ and $\tau_k \in [0, 0.018]s$.

 Fig. 4. Trajectories of $x_2(t)$ and $x_2(t_k)$ with $h = 0.02s$ and $\tau_k \in [0, 0.018]s$.

A. Example 1: Mass Spring System

Consider the following nonlinear mass-spring system [36]:

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= -0.01x_1 - 0.67x_1^3 + w + u\end{aligned}\quad (35)$$

where $x_1 \in [-1, 1]$ and $w = 0.2\sin(2\pi t)\exp(-t)$ is external disturbance.

Choose fuzzy membership function as $\mu_1(x_1) = 1 - x_1^2$ and $\mu_2(x_1) = 1 - \mu_1(x_1)$. The following fuzzy model is used to model the nonlinear system:

Rule 1 :

IF x_1 is μ_1 , THEN $\dot{x} = A_1x + B_1u + B_{11}w$
 $z = C_1x + D_1u$

Rule 2 :

IF x_1 is μ_2 , THEN $\dot{x} = A_2x + B_2u + B_{12}w$
 $z = C_2x + D_2u$ (36)

where $A_1 = \begin{bmatrix} 0 & 1 \\ -0.01 & 0 \end{bmatrix}$, $A_2 = \begin{bmatrix} 0 & 1 \\ -0.68 & 0 \end{bmatrix}$, $B_1 = B_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $B_{11} = B_{12} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $C_1 = C_2 = [0 \ 1]$, and $D_1 = D_2 = 0$.

We set $\delta = 0.2s$, $\varepsilon_{ij} = 10$ ($1 \leq i \leq j \leq 2$). Then applying Theorem 2 and Remark 3 yields a locally optimal combination: $\epsilon_1 = 1.1346$, $\epsilon_2 = 1.3285$, $\epsilon_3 = 1.4287$, and $\epsilon_4 = 0.5746$. At the same time, we can obtain $\gamma_{min} = 1.2448$ by solving a linear objective minimization problem (29), and controller gain matrices $K_1 = [-1.7458 \ -3.1297]$, $K_2 = [-1.1205 \ -3.1230]$.

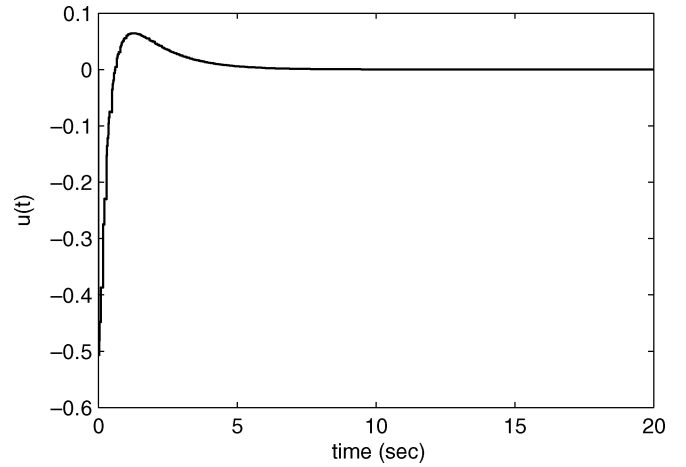
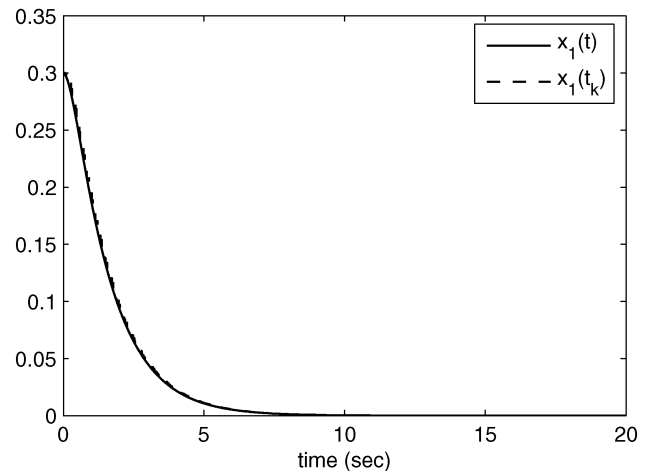
Next, under the same initial value $x(0) = [0.3, 0]^T$, we show the simulation results with different sampling period and packet dropout rate.

CASE I: $h = 0.02s$, $\tau_k \in [0, 0.018]s$.

The state and control input trajectories of (35) are shown in Figs. 3–5, respectively. The system can be stabilized well. Note that no packet dropout occurs in this simulation case, while the maximum allowable packet dropout rate r_{max} is 88.8% derived by (6). Therefore, the sampling period h can be increased properly.

CASE II: $h = 0.05s$, $\tau_k \in [0, 0.045]s$.

The state and control input trajectories of (35) are shown in Figs. 6, 8, and 9, respectively. Moreover Fig. 7 is the local part of Fig. 6 in time interval $[3, 3.5]$. The effect of random network-induced delay is shown clearly in this figure. There is also


 Fig. 5. Control input $u(t)$ of the NCS with $h = 0.02s$ and $\tau_k \in [0, 0.018]s$.

 Fig. 6. Trajectories of $x_1(t)$ and $x_1(t_k)$ with $h = 0.05s$ and $\tau_k \in [0, 0.045]s$.

no packet dropout in this simulation case, while the maximum allowable packet dropout rate r_{max} is 66.7% derived by (6).

Comparing **CASE I** and **CASE II**, there are nothing obviously different from the figures. The computational time of CPU is approximately 51.1936 s in **CASE I**, but only 19.4179 s in **CASE II** on a computer with a 1.2-GHz processor and 256 MB of

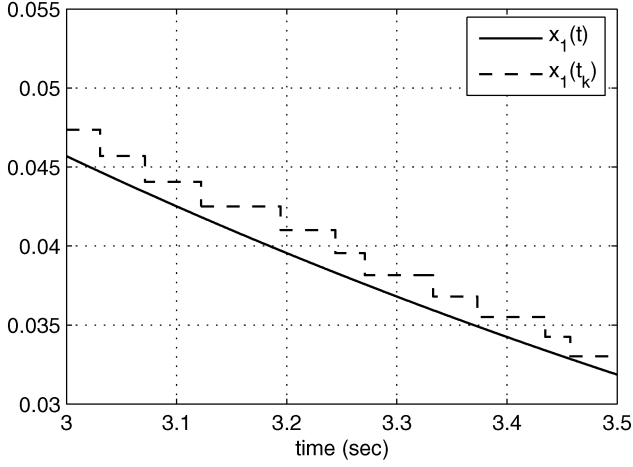


Fig. 7. Trajectories of $x_1(t)$ and $x_1(t_k)$ with $h = 0.05$ s and $\tau_k \in [0, 0.045]$ s.

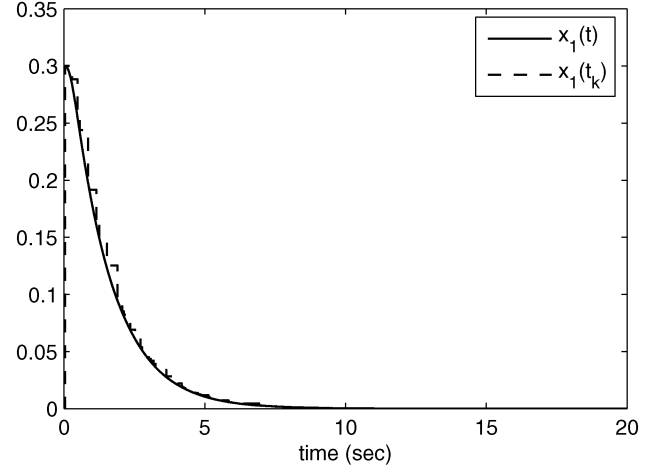


Fig. 10. Trajectories of $x_1(t)$ and $x_1(t_k)$ with $h = 0.05$ s, $\tau_k \in [0, 0.045]$ s, and $r = 60\%$.

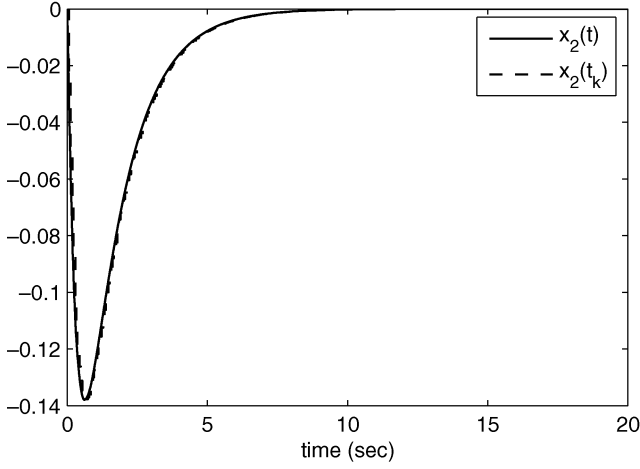


Fig. 8. Trajectories of $x_2(t)$ and $x_2(t_k)$ with $h = 0.05$ s and $\tau_k \in [0, 0.045]$ s.

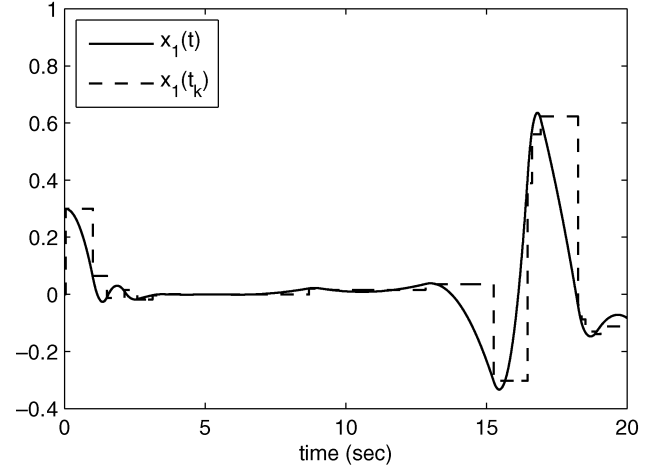


Fig. 11. Trajectories of $x_1(t)$ and $x_1(t_k)$ with $h = 0.05$ s, $\tau_k \in [0, 0.045]$ s, and $r = 90\%$.

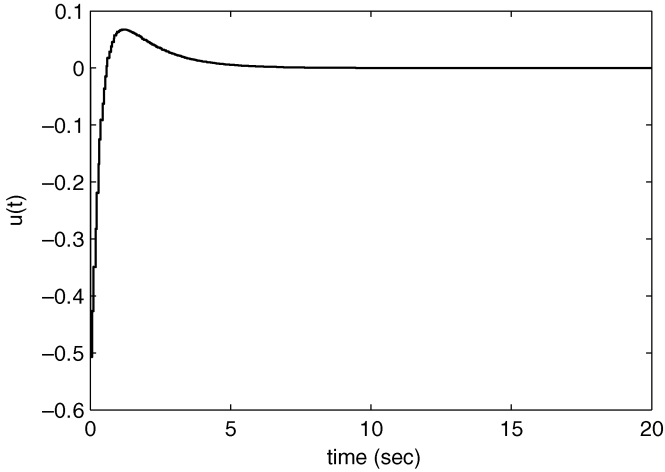


Fig. 9. Control input $u(t)$ of the NCS with $h = 0.05$ s and $\tau_k \in [0, 0.045]$ s.

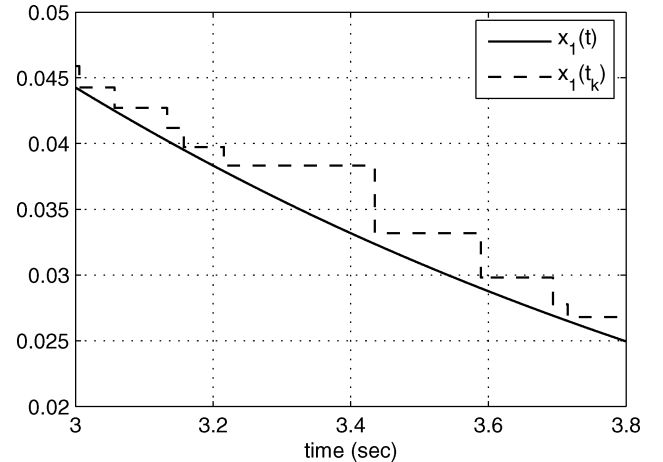


Fig. 12. Trajectories of $x_1(t)$ and $x_1(t_k)$ with $h = 0.05$ s, $\tau_k \in [0, 0.045]$ s, and $r = 60\%$.

memory for running the MATLAB program. So the communication burden of *CASE II* is much less than that of *CASE I*.

CASE III: $h = 0.05$ s, $\tau_k \in [0, 0.045]$ s with different packet dropout rate.

We try to explore how the packet dropout affects the stability of the NCS. For $\delta = 0.2$ s, the maximum allowable packet dropout rate r_{max} is 66.7%. So we simulate packet dropout phenomena with $r = 60\%$ and $r = 90\%$ to verify our results. The

simulation results are shown in Figs. 10 and 11, respectively. Comparing Fig. 6 with Fig. 10, the performance of the system does not degrade much, but the system is unstable in Fig. 11. Moreover, Fig. 12 is the local part of Fig. 10 in time interval [3, 3.8], where the trajectory of $x(t_k)$ indicates that two sequential control signals arrive not more than 0.2 s. It illustrates if a transmission of packet takes place at instant t_k and the control signal

will reach the plant at the instant $t_k + \tau_k$, then the next control signal must arrive within the time interval $(t_k, t_k + \delta]$. Therefore, the maximum allowable control interval δ can be used to reflect both the network-induced delay and packet dropout problems. Moreover, it is important to choose a proper sampling period h to satisfy $r_{max} > r$ based on δ when packets dropout occur inevitably in real network condition.

B. Example 2: Single-Link Rigid Robot System

Consider the following nonlinear system of a single-link rigid robot connected through a revolute joint to the basement and whose plane of motion is vertical, which is controlled through a network. The motion equation of this mechanical system is given by [26]

$$J\ddot{\vartheta} = -(0.5 mgl + Mgl)\sin(\vartheta) + u \quad (37)$$

where ϑ denotes the joint rotation angle in radians, $m = 1.5$ kg is the mass of the load, $M = 3$ kg is the mass of the rigid link, $g = 9.8$ m/s² is the gravity constant, $l = 0.5$ m is the length of the robot link, $J = Ml^2 + (1/3)ml^2 = 0.875$ kg · m² is the moment of inertia, and u is the control torque applied at the joint in Nm. $\vartheta = 0$ denotes the lowest vertical equilibrium position under zero control torque. The control task is to move the robot arm from any initial state $\vartheta \in [0, (\pi/2)]$ to the equilibrium position define by $\vartheta = 0$, $\dot{\vartheta} = 0$, and $\ddot{\vartheta} = 0$ in spite of perturbations of plant parameters in J .

Define new state variables as $x_1 = \vartheta$, $x_2 = \dot{\vartheta}$. Applying the approach in [31], we can construct a T-S fuzzy model for system (37) as follows:

Rule 1 :

IF x_1 is about 0, THEN $\dot{x} = (A_1 + \Delta A_1)x + B_1u + w$

Rule 2 :

IF x_1 is about $\pi/2$, THEN $\dot{x} = (A_2 + \Delta A_2)x + B_2u + w$ (38)

where

$$A_1 = \begin{bmatrix} 0 & 1 \\ \frac{-(0.5 mgl + Mgl)}{J} & 0 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 1 \\ \frac{-2(0.5 mgl + Mgl)}{\pi J} & 0 \end{bmatrix}$$

$$B_1 = B_2 = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix},$$

and the matrices ΔA_i ($i = 1, 2$) satisfies condition (2). For simplicity, we choose

$$G = I, \quad E_{a1} = \begin{bmatrix} 0 & 0 \\ 0.5 & 0 \end{bmatrix}, \quad \text{and} \quad E_{a2} = \begin{bmatrix} 0 & 0 \\ -0.5 & 0 \end{bmatrix}. \quad (39)$$

The fuzzy membership functions are set as $\mu_1(x_1) = (0.5\pi - |x_1|)/0.5\pi$, and $\mu_2(x_1) = 1 - \mu_1(x_1)$. In this example, the controlled output equation is assumed as

$$z = Cx \quad C = [0 \quad 1].$$

We set $\delta = 0.15s$, $\varepsilon_{ij} = 10$ ($1 \leq i \leq j \leq 2$). Then applying Theorem 2 and Remark 3 yields a locally optimal combination: $\varepsilon_1 = 1.1836$, $\varepsilon_2 = 1.1773$, $\varepsilon_3 = 2.0776$, and $\varepsilon_4 = 0.5551$. Moreover, we can obtain $\gamma_{min} = 1.1256$ by solving (29), and controller gain matrices

$$K_1 = [10.5000 \quad -5.1057]$$

$$K_2 = [3.8800 \quad -5.0628]. \quad (40)$$

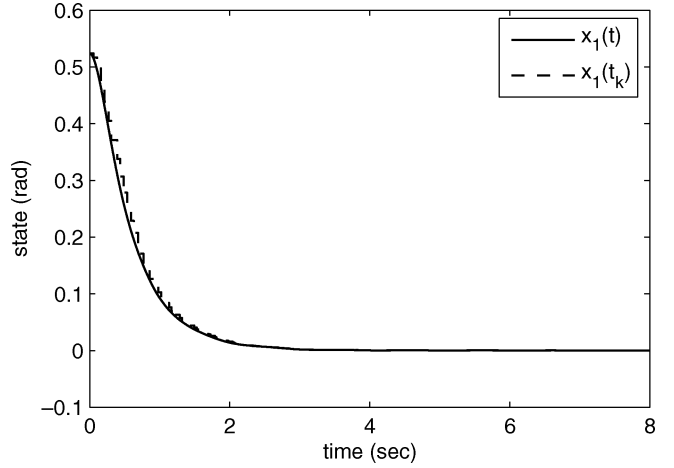


Fig. 13. Trajectories of $x_1(t)$ and $x_1(t_k)$ with $h = 0.05s$ and $\tau_k \in [0, 0.045]s$.

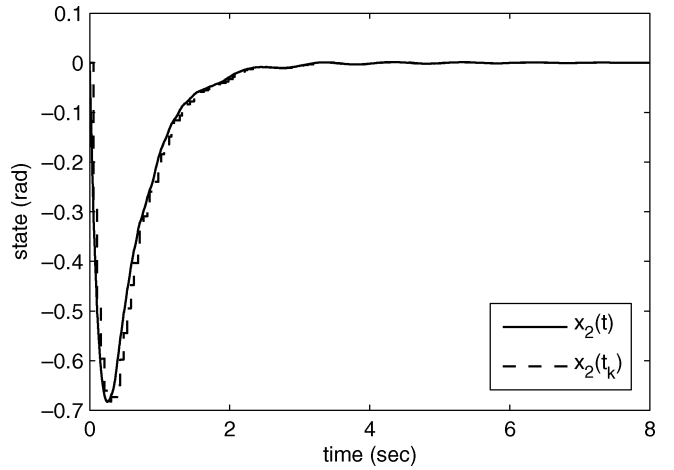


Fig. 14. Trajectories of $x_2(t)$ and $x_2(t_k)$ with $h = 0.05s$ and $\tau_k \in [0, 0.045]s$.

If the initial conditions are chosen as $x(0) = [\pi/6, 0]^T$, the external disturbance signal is chosen as $w = 0.5 \sin(2\pi t) \exp(-0.5t)$, the sampling period and network-induced delay are chosen as $h = 0.05s$ and $\tau_k \in [0, 0.045]s$, respectively, the simulation results for the two states of the nonlinear NCS (37) are shown in Figs. 13 and 14, from which one can clearly see the effectiveness of the proposed design method. From Fig. 15, it is easy to see that the given T-S model satisfies $\|z\|_2 < \gamma\|w\|_2$ under the zero initial condition and controller gains (40) for all $t \geq 0$.

It is detected that the given T-S model is robustly stabilizable under controller (8) with controller gains (40) and satisfies $\|z\|_2 < \gamma\|w\|_2$ under zero initial condition for any nonzero $w \in \mathcal{L}_2$ and all admissible uncertainties satisfying (39) in the presence of network.

C. Example 3: CSTR System

Consider a CSTR system in which the first-order irreversible exothermic reaction occurs. The mass balance and energy balance equations in a well-stirred region uniform in composition C and temperature T are governed by [44]

$$V_{ol} \frac{dC}{dt} = \lambda(C_0 - C) - V_{ol}R$$

$$V_{ol}C_p \frac{dT}{dt} = \lambda(T_0 - T) + (\Delta H)V_{ol}R$$

$$- U(T - T_w + h(t)) \quad (41)$$

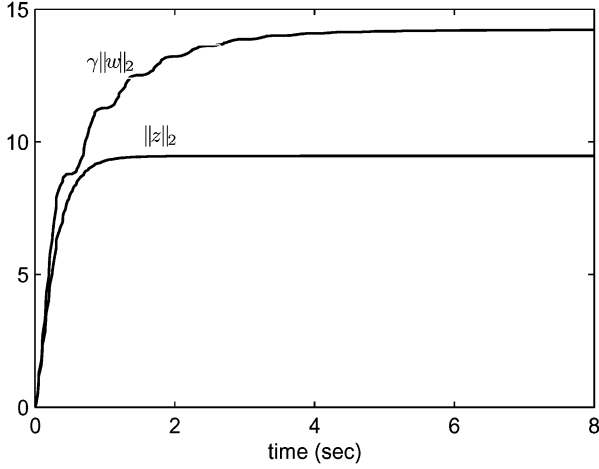


Fig. 15. Demonstration for the inequality $\|z\|_2 < \gamma\|w\|_2$ under the zero initial condition.

respectively, where V_{ol} is the constant reaction volume, λ is the flow rate in and out, C_0 is the feed concentration of the component of interest, R is the reaction rate per unit volume, C_p is the heat capacity per unit volume of the flowing materials as well as the contents of the reactor, T_0 is the feed temperature, ΔH is the molar heat of reaction, taken to be positive for exothermic systems, U is the overall heat transfer coefficient, in units which include the area of transfer surface, T_w is coolant or heating fluid temperature, and $h(t)$ represents an external disturbance, which may be caused by the uncontrollable change in the ambient temperature.

Assume the first-order kinetics for reaction rate

$$R = \kappa_0 C \exp\left(\frac{-Q}{T}\right)$$

where κ_0 is the reaction velocity constant, Q is the ratio of an Arrhenius activation energy to the gas constant. By using variable transformation as follows:

$$x_1 = \frac{C_0 - C}{C_0}, \quad x_2 = \frac{T - T_0}{Q}$$

$$\nu = \frac{t}{\frac{V_{ol}}{\lambda}}, \quad u = \frac{T_w - T_0}{Q}$$

we can rewrite (41) as

$$\begin{aligned} \frac{dx_1}{d\nu} &= -x_1 + D_a(1 - x_1) \exp\left(-\frac{1}{x_2 + \rho}\right) \\ \frac{dx_2}{d\nu} &= -(1 + \varrho)x_2 + HD_a(1 - x_1) \\ &\quad \times \exp\left(-\frac{1}{x_2 + \rho}\right) + \varrho(u + w) \end{aligned} \quad (42)$$

where

$$D_a = \frac{\kappa_0 V_{ol}}{\lambda}, \quad H = \frac{(\Delta H)C_0}{QC_p}$$

$$\rho = \frac{T_0}{Q}, \quad \varrho = \frac{U}{\lambda C_p}, \quad w = -\frac{h(t)}{Q}.$$

For a given group of parameters D_a , H , ρ , and ϱ , (42) might have one, two, or three equilibrium points corresponding to different steady input $u = u_e$. Choosing parameters as

$$D_a = 0.072, \quad H = 8, \quad \rho = 20, \quad \varrho = 0.3$$

and letting $u_e = 0$, we can get the system (42) has only one equilibrium point, which is

$$x_e = \begin{bmatrix} 0.0642 \\ 0.3948 \end{bmatrix}.$$

Define new state variables as

$$\iota x_1 = x_1 - x_{e1}$$

$$\iota x_2 = x_2 - x_{e2}.$$

Then, the dynamical model of (42) can be rewritten as

$$\begin{aligned} \frac{d\iota x_1}{d\nu} &= -\iota x_1 - x_{e1} + D_a(1 - \iota x_1 - x_{e1}) \\ &\quad \times \exp\left(-\frac{1}{\iota x_2 + x_{e2} + \rho}\right) \\ \frac{d\iota x_2}{d\nu} &= -(1 + \varrho)(\iota x_2 + x_{e2}) + HD_a(1 - \iota x_1 - x_{e1}) \\ &\quad \times \exp\left(-\frac{1}{\iota x_2 + x_{e2} + \rho}\right) + \varrho(u + w). \end{aligned} \quad (43)$$

Assume that there is some perturbation in H because the parameter H is difficult to measure exactly. In addition, the dynamics of the temperature is readily influenced by the change of external environmental temperature, which can be modeled by a time-varying disturbance. Therefore, the controlled output equation is given as

$$z = \begin{bmatrix} 0 & 1 \end{bmatrix} x = \begin{bmatrix} 0 & 1 \end{bmatrix} (\iota x + x_e).$$

In this example, we assume that

$$w(\nu) = 0.5 \cos(2\pi\nu) \exp(-\nu).$$

In short, our aim is to design the robust H_∞ controller for system (43) in network-based environment described in Fig. 1. So we need to use T-S fuzzy model to represent system (43) firstly. According to the linearization of (43) around the points $\iota x^1 = [0 \ -1]^T$, $\iota x^2 = [0 \ 0]^T$, and $\iota x^3 = [0 \ 1]^T$, respectively, we have the following three fuzzy rules [44]:

Rule 1 : IF ιx_2 is small (i.e. ιx_2 is about -1), THEN

$$\frac{d\iota x}{d\nu} = (A_1 + \Delta A_1)\iota x + B_1(u + w)$$

Rule 2 : IF ιx_2 is middle (i.e. ιx_2 is about 0), THEN

$$\frac{d\iota x}{d\nu} = (A_2 + \Delta A_2)\iota x + B_2(u + w)$$

Rule 3 : IF ιx_2 is large (i.e. ιx_2 is about 1), THEN

$$\frac{d\iota x}{d\nu} = (A_3 + \Delta A_3)\iota x + B_3(u + w) \quad (44)$$

where

$$A_1 = \begin{bmatrix} -1.0684 & 0.0002 \\ -0.5471 & -1.2987 \end{bmatrix}, \quad A_2 = \begin{bmatrix} -1.0686 & 0.0002 \\ -0.5484 & -1.2988 \end{bmatrix}$$

$$A_3 = \begin{bmatrix} -1.0687 & 0.0001 \\ -0.5497 & -1.2988 \end{bmatrix}, \quad B_1 = B_2 = B_3 = \begin{bmatrix} 0 \\ 0.3 \end{bmatrix}.$$

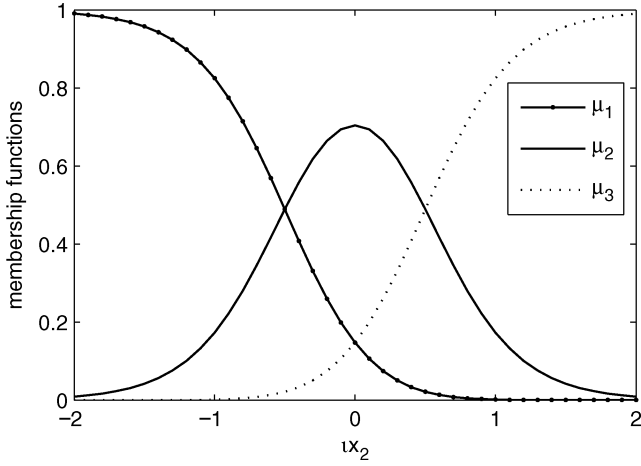


Fig. 16. Membership function of CSTR fuzzy model.

Suppose that there is not more than 5% perturbation in H and the matrices ΔA_i ($i = 1, 2, 3$) satisfy condition (2). For simplicity, we choose

$$G = I, \quad Ea_1 = Ea_2 = Ea_3 = \begin{bmatrix} 0 & 0 \\ -0.0274 & -0.0649 \end{bmatrix}.$$

The membership functions are selected as

$$\begin{aligned} \mu_1 &= \frac{\exp\left(\frac{-(u_{x_2}+1)^2}{0.8^2}\right)}{\exp\left(\frac{-(u_{x_2}+1)^2}{0.8^2}\right) + \exp\left(\frac{-(u_{x_2})^2}{0.8^2}\right) + \exp\left(\frac{-(u_{x_2}-1)^2}{0.8^2}\right)} \\ \mu_2 &= \frac{\exp\left(\frac{-(u_{x_2})^2}{0.8^2}\right)}{\exp\left(\frac{-(u_{x_2}+1)^2}{0.8^2}\right) + \exp\left(\frac{-(u_{x_2})^2}{0.8^2}\right) + \exp\left(\frac{-(u_{x_2}-1)^2}{0.8^2}\right)} \\ \mu_3 &= \frac{\exp\left(\frac{-(u_{x_2}-1)^2}{0.8^2}\right)}{\exp\left(\frac{-(u_{x_2}+1)^2}{0.8^2}\right) + \exp\left(\frac{-(u_{x_2})^2}{0.8^2}\right) + \exp\left(\frac{-(u_{x_2}-1)^2}{0.8^2}\right)} \end{aligned}$$

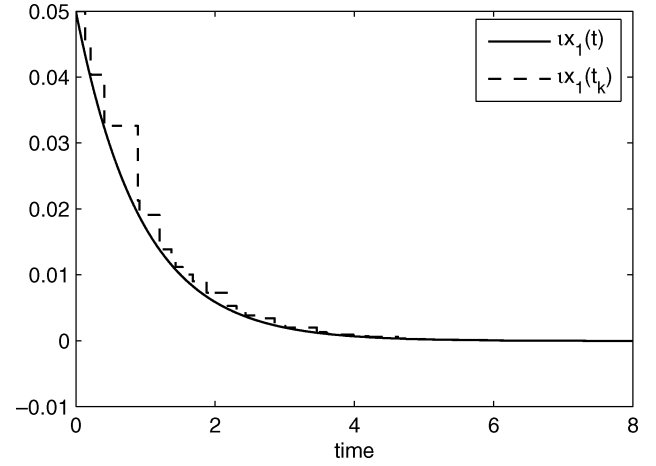
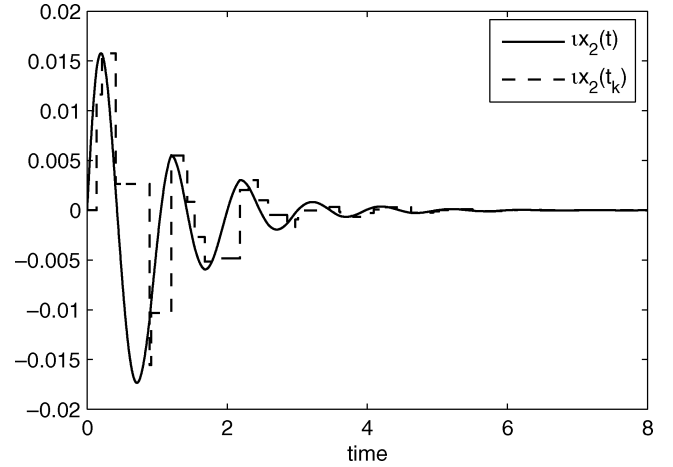
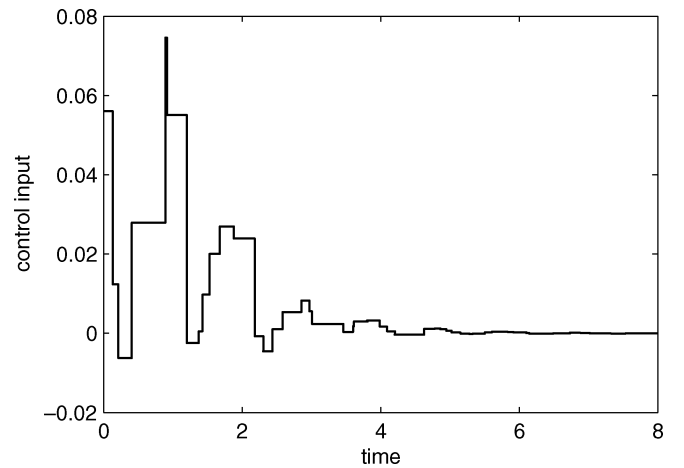
which are shown in Fig. 16.

Next, using Theorem 2 and optimization problem (29) with $\epsilon_1 = 1$, $\epsilon_2 = 1.2$, $\epsilon_3 = 1.2$, and $\epsilon_4 = 0.6$, for $\delta = 0.5s$, $\varepsilon_{ij} = 20$ ($1 \leq i \leq j \leq 3$), we obtain $\gamma_{min} = 0.1654$, and controller gain matrices $K_1 = [1.1181 \quad -3.2636]$, $K_2 = [1.1197 \quad -3.2632]$, and $K_3 = [1.1213 \quad -3.2633]$.

Figs. 17–19 show the trajectories of $u_x(t)$ and $u_x(t_k)$ and control signal of the NCS, respectively, with initial conditions $u_x(0) = [0.05, 0]^T$, sampling period $h = 0.1$, network-induced delay $\tau_k \in [0, 0.12]$, and packet dropout rate $r = 50\%$. In this example, simulation shows that the NCS is stabilized with sub-optimal H_∞ performance. Also, control signal is a piecewise constant function explicitly from Fig. 19. Network-induced delays τ_k ($k = 0, 1, 2, \dots$) are random and can exceed the sampling period as shown in Fig. 20. In fact, we do not constrain the size of network-induced delays in our results. We need only condition (4) to be satisfied. In addition, the H_∞ performance in this example lies in attenuating not only external disturbance but also the modeling error between the fuzzy model (44) and the nonlinear system (43).

V. CONCLUSION

In this paper, a robust H_∞ control method has been proposed for a class of nonlinear NCSs, which can be represented


 Fig. 17. Trajectories of $u_{x_1}(t)$ and $u_{x_1}(t_k)$ with $h = 0.1$, $\tau_k \in [0, 0.12]$, and $r = 50\%$.

 Fig. 18. Trajectories of $u_{x_2}(t)$ and $u_{x_2}(t_k)$ with $h = 0.1$, $\tau_k \in [0, 0.12]$, and $r = 50\%$.

 Fig. 19. Control input of the NCS with $h = 0.1$, $\tau_k \in [0, 0.12]$, and $r = 50\%$.

by a T-S fuzzy model with uncertainties. The developed controller can be obtained by solving a set of LMIs based on a Lyapunov–Krasovskii functional method. In the proposed approach, the maximum allowable control interval δ is adopted to reflect both the network-induced delay and packet dropout problems. Moreover, the network-induced delay τ_k needs not

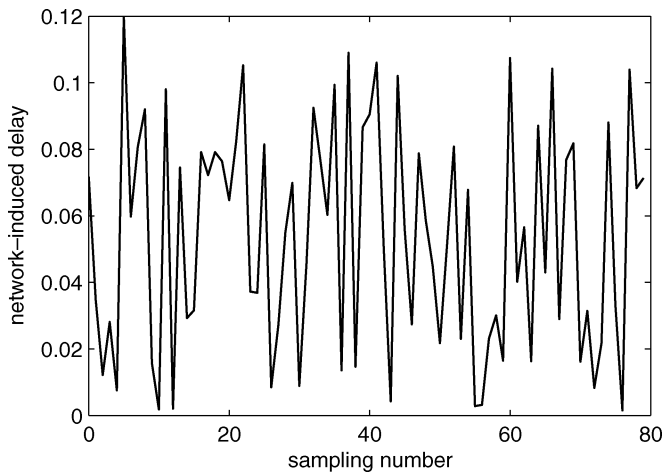


Fig. 20. Network-induced delay τ_k .

to be restricted within one sampling period. Simulation results of three illustrative examples show the validity of the proposed method.

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